

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 17/ 05/ 2019

Day: Friday

Time: 10:00 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : B.Optomtry May-2019
Semester: 3 rd
Section : I
Course Code : BOP302
Course Name: Visual optics-I

Important Instructions:

1. There are two sections in the paper.
2. **SECTION –I** contains 2 Questions:
 - a. MCQs for 20 marks answered in the question paper itself, should be submitted within 30 minutes.
 - b. Short answers for 20 marks (to be answered in separate answer book)
3. **SECTION – II** contains only 2 Question of Short notes and Essays (to be answered in answer book)
4. Marks on the right side suggest the total marks of that question.
5. Maximum time allotted for both section is 3 hours.
6. Candidates who complete **SECTION – I** (MCQs) earlier than 30 minutes can collect main question paper immediately after handing over the **SECTION-I (MCQs)** question paper with answers.
7. Non-programmable calculators are only allowed.
8. Draw the figures where necessary.

SECTION-I

Q.1. Multiple Choice Questions: (MCQs) (1 X 10= 10)

Encircle the correct answer

1. **Rod cells are responsible for**_____.
 - a. Day light vision
 - b. Night vision
 - c. Colour vision
 - d. None of the above
2. **Pelli-robson chart is used to test**_____.
 - a. Colour vision
 - b. Visual acuity
 - c. Contrast sensitivity
 - d. Night vision

3. _____ is two position keratometer.
- a. Bausch & Lomb
 - b. Javal schiotz
 - c. Auto keratometer
 - d. None of the above
4. **Bending of light rays when it passes from one medium to another medium is called as_____**
- a. Diffraction
 - b. Refraction
 - c. Reflection
 - d. Total internal reflection
5. **Lacquer's cracks are seen in case of_____.**
- a. Hypermetropia
 - b. Myopia
 - c. Both a & b
 - d. None of the above
6. **Deuteranopia means_____.**
- a. Absence of medium wavelength sensitive cone cells
 - b. Absence of long sensitive cone cells
 - c. Absence of short sensitive cone cells
 - d. None of the above
7. _____ can be used to check visual acuity in infants and toddlers.
- a. Fix and follow
 - b. OKN drum
 - c. Cardiff acuity cards
 - d. All of the above
8. _____ is responsible for accommodation.
- a. Tear film
 - b. Lens
 - c. Cornea
 - d. Retina
9. **Astigmatism leads to formation of_____**
- a. Focused image from two meridians on single point
 - b. Focal lines
 - c. Circle of Least confusion
 - d. Both b & c.
10. **Headache is commonest in case of_____.**
- a. Uncorrected Myopia
 - b. Uncorrected hyperopia
 - c. Aphakia
 - d. None of the above

Identify the sentence as True or False:

(5 × 1 = 5)

11. Ishihara test works on colour arrangement principle.
12. Myopic fundus shows chorioretinal atrophy
13. Accommodative esotropia can happen in case of hyperopia
14. Plus lens produces barrel shaped distortion
15. Coma is a chromatic aberration.

Match the following:

(5 × 1 = 5)

- | | |
|------------------------|---------------------------------|
| 16. Sine wave gratings | A. 5 Optotypes |
| 17. Panel D15 | B. Myopic fundus |
| 18. LOGMAR Chart | C. Colour vision |
| 19. Crescent | D. Present in peripheral retina |
| 20. Rod cells | E. Contrast sensitivity |

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B.Optomety Examination
May 2019

BOP302: Visual optics-I

Date: 17/05/2019 Day: Friday Time: 10:30am to 1:00pm Maximum Marks: 50

Instructions:

1. There are two sections in the paper.
 2. **SECTION –I** contains 2 Questions:
 - a. MCQs for 20 marks answered in the question paper itself, should be submitted within 30 minutes.
 - b. Short answers for 20 marks (to be answered in separate answer book)
 3. **SECTION – II** contains only 2 Question of Short notes and Essays (to be answered in answer book)
 4. Marks on the right side suggest the total marks of that question.
 5. Maximum time allotted for both section is 3 hours.
 6. Candidates who complete **SECTION – I (MCQs)** earlier than 30 minutes can collect main question paper immediately after handing over the **SECTION-I (MCQs)** question paper with answers.
 7. Non-programmable calculators are only allowed.
 8. Draw the figures where necessary.
-

SECTION-I

Q.2. Short answers (10 out of 11)

(10 × 2 = 20)

- a. What is the use of Vistech chart?
- b. Enlist two fundus changes which can be seen in case of pathological myopia.
- c. Enlist two Use of Keratometer.
- d. Define Astigmatism.
- e. What is Dark adaptation?
- f. What is Protanomaly?
- g. Enlist the names of two visual acuity charts which can be used to check visual acuity in case of illiterate patients.
- h. Write two symptoms of hyperopic patient.
- i. What is vergence?
- j. What is catoptric power?
- k. What is emmetropization?

SECTION II

Q.1. Short Notes (4 out of 5)

(4 × 5 = 20)

- a. Write a note on light and dark adaptation.
- b. Write a note on Trichromatic theory of colour vision.
- c. Write a note on OKN drum.
- d. Draw labeled diagram of reduced eye.
- e. Draw optical train of Bausch & Lomb keratometer.

Q.2. Essay (1 out of 2)

(1 × 10 = 10)

- a. Define hyperopia. Classify it according to action of accommodation with example
- b. Write a note on Contrast sensitivity.